

OLLSCOIL NA hÉIREANN, CORCAIGH
THE NATIONAL UNIVERSITY OF IRELAND, CORK

COLÁISTE NA hOLLSCOILE, CORCAIGH
UNIVERSITY COLLEGE, CORK

Examination Session and Year	Winter 2022
Module Code	AC2204
Module Title	INTRODUCTION TO MANAGEMENT ACCOUNTING
Paper Number	1
External Examiner	Professor Martin Quinn
Head of the Department	Professor Mark Mulcahy
Internal Examiners	Dr Margaret Healy
Instructions to Candidates	Answer three questions in total. Please attempt QUESTION ONE from Section A and TWO questions from Section B.
Duration of Paper	1.5 hours
Special Requirements	Show your workings; marks will be awarded for workings. Students are allowed to use calculators that have standard arithmetic, scientific, graphical or algebraic functions provided they are not programmable and can not store text (other than algebraic notation).

PLEASE DO NOT TURN THIS PAGE UNTIL INSTRUCTED TO DO SO

THEN ENSURE THAT YOU HAVE THE CORRECT EXAM PAPER

SECTION A: COMPULSORY

QUESTION ONE

The owner of Grove Ltd have asked for your advice and expertise in determining profit expectations. The company currently sells a single product, garden benches, for which the following budgeted financial data for 2023 is available.

Sales price per bench (unit)	€240
Variable cost per bench (unit)	€150
Total fixed expenses	€5,400

Required:

(A) Assume 75 units (benches) will be sold in 2023. Based on the information provided, calculate each of the following and explain their usefulness for financial planning at Grove Ltd.

- (i) Operating profit (loss);
- (ii) Breakeven level of sales in units;
- (iii) Breakeven level of revenues (using the contribution margin approach);
- (iv) Percentage margin of safety;
- (v) Operating leverage.

(20 marks)

(B) Grove Ltd is considering a proposal to expand by launching a second product, Picnic tables. Initial market research suggests there would be strong demand for the product, with each unit selling for €500. Picnic tables are relatively easy to manufacture, with variable cost of €100 per unit projected. If the proposal goes ahead, additional marketing costs of €4,010 per annum will be incurred and Grove Ltd will need to rent a storage unit for raw materials at a cost of €6,000 per annum. Grove Ltd expects to sell 1 “Picnic table” unit for every 3 “Garden bench” units sold.

- (i) Using the Weighted Average Contribution Margin (WACM) approach, determine the breakeven point in units and sales revenues for Grove Ltd under the expansion proposal.
- (ii) If actual sales are 75 “Garden bench” units and 30 “Picnic table” units, determine the expected operating profit (loss) from proceeding with the expansion proposal.
- (iii) Advise Grove Ltd on whether or not they should proceed with the proposal, giving one financial and one non-financial reason for your answer.

(14 marks)

(C) Identify and discuss two assumptions underlying CVP analysis. In your answer, consider the extent to which they can be held as reasonable approximations of reality.

(6 marks)

(Total: 40 marks)

SECTION B
ANSWER ANY TWO QUESTIONS

QUESTION TWO

Fraser Ltd commenced operations on Jan 1st 2020 manufacturing and selling a single product from a premises in Cork city. The following information is available for the year to 31st December 2020.

Sales (in units)	3,600
Selling price per unit	€26.00
Production (in units)	4,000
Variable manufacturing expenses (total)	€49,200
Fixed manufacturing expenses (total)	€16,800
Variable selling expenses (total)	€14,400
Fixed selling expenses (total)	€11,000

Required:

- (A)** Calculate the unit product cost, the value of closing stock and the operating profit using each of the following methods:

- (i)** Marginal costing;
- (ii)** Absorption costing.

[Please show the income statement for each method, as part of your answer]

(18 marks)

- (B)** Management at Fraser Ltd is undecided about choosing between marginal costing and absorption costing. Discuss two arguments in support of the use of each of the methods.

(12 marks)

(Total: 30 marks)

QUESTION THREE

Diver Ltd uses a traditional (absorption) costing system to determine the cost of each of its three products. Production overheads (totalling €1,232,000) are absorbed on the basis of machine hours, at a rate of €28 per machine hour. Direct labour costs €18 per hour. The following additional data is available for the most recent financial period.

	Total production volume (units)	Direct Labour hours per unit	Machine hours per unit	Total machine hours	Direct Material cost per unit
Product O	10,000	3.0 hours	3.0 hours	30,000	€20
Product P	1,000	1.0 hours	2.0 hours	2,000	€10
Product S	3,000	1.0 hours	4.0 hours	12,000	€16

An activity-based costing system is under consideration at Diver Ltd. The production manager has estimated that production overheads can be divided as shown below.

Cost Pool	% of total overhead	Cost Driver
Machining	40%	No. of machine hours
Production scheduling	30%	No. of set-ups
Quality control	20%	No. of units
Materials handling	10%	No. of purchase orders

Activity volumes for the financial period under review are detailed below.

	No. of purchase orders	No. of set-ups
Product O	10	50
Product P	10	20
Product S	5	50

Required:

- (A) Calculate the total costs and the cost per unit for each product using:
- (i) Traditional (absorption) costing, with production overheads allocated based on machine hours.
 - (ii) Activity-based costing, with production overheads allocated based on the cost drivers given in the question.

(22 marks)

- (B) Explain each of the following terms used in management accounting and outline their impact on the cost per unit of a product.
- (i) Product-cost cross-subsidization.
 - (ii) Batch-level activities.

(8 marks)

(Total: 30 marks)

QUESTION FOUR

Consider each of the following statements. For each statement, state whether you agree or disagree, explaining your answer in each case.

- (A) Sunk costs quantify the potential benefit that is given up when one alternative is selected over another.
- (B) Pigeon Ltd's manufacturing overhead is 40% of its total conversion costs. If direct labour is €63,000 and direct materials are €45,860, then the manufacturing overhead is €42,000.
- (C) The following are activity levels and associated costs for Pamela's Pet Parlour for 2022.

Low:	April	150 dogs groomed	Total cost: €5,320
High:	August	800 dogs groomed	Total cost: €11,040

Then, if it is planned to groom 200 dogs in January 2023, based on the cost formula for 2022, the expected cost is €3,444.

- (D) If a given cost is direct with respect to one stated cost objective, then it must be direct with respect to all other cost objectives.
- (E) Cost structure refers to the relative proportions in which an organisation's products are sold.

(5 * 6 marks each)
(Total: 30 marks)

**PLEASE DO NOT TURN THIS
PAGE UNTIL INSTRUCTED TO
DO SO**

**THEN ENSURE THAT YOU
HAVE THE CORRECT EXAM
PAPER**